Graph-Based Framework for Identifying Contradictions and Circular Reasoning in Debates

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Guided By,

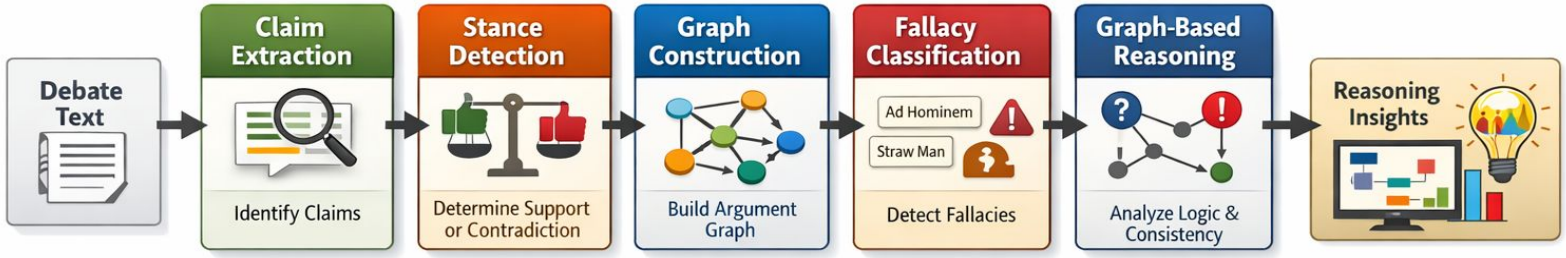
Dr. Emy Mariam George

End-to-End Pipeline Overview

The proposed framework follows a sequential pipeline that transforms raw debate text into structured reasoning insights. Each module is independently evaluable while contributing to the overall system.

1. **Claim Extraction** - Identify atomic argumentative claims from raw text.
2. **Stance Detection** - Determine support, contradiction, or neutrality between claims.
3. **Graph Construction** - Represent claims and relations as directed graphs.
4. **Fallacy Classification** - Assign claim-level fallacy labels.
5. **Graph-Based Reasoning** - Detect contradictions, cycles, and weak arguments.
6. **Visualization & Analysis** - Present interpretable reasoning paths

This modular design enables targeted optimization and benchmarking of each stage.



Claim Extraction Module

The first stage of the pipeline focuses on extracting meaningful argumentative units from raw debate text. In the current implementation, sentence-level segmentation is used as a baseline approach to identify candidate claims.

Using the Natural Language Toolkit (NLTK), debate text is segmented into grammatically complete sentences. These sentences are treated as provisional claims and serve as nodes in the stance graph.

Observations

- Sentence-level extraction reliably captures explicit claims in structured debates.
- However, complex claims spanning multiple sentences or implicit premises are not fully captured.
- This limitation motivates future integration of supervised argument-mining models.

Results

- Produced clean node sets for downstream stance detection.

Stance Detection Using NLI Models

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Stance detection determines whether one claim supports, contradicts, or is neutral with respect to another claim. This task is modeled as a Natural Language Inference (NLI) problem.

Each ordered claim pair (c_i, c_j) is encoded using the RoBERTa-large-MNLI format, where one claim acts as the premise and the other as the hypothesis. The model outputs probabilities for entailment, contradiction, and neutrality.

To ensure reliability, only predictions above a confidence threshold (0.75) are retained as graph edges.

Model Configuration

- Model: RoBERTa-large-MNLI
- Hidden size: 1024
- Output classes: support, contradict, neutral

Stance Detection Performance and Benchmark

The stance detection module was evaluated on debate-oriented datasets and compared against established benchmarks.

Model	Dataset	Accuracy / Reliability
BERT-NLI	multi_nli	~74%
RoBERTa-MNLI	multi_nli	~81%
GraphNLI (Paper)	Kialo	~86%
Our Filtered Pipeline	**syncIALO**	**~64%**

Analysis

- Confidence-based pruning significantly reduces noisy edges.
- Errors are primarily due to implicit reasoning, no context clues and long-range dependencies.

Stance Graph Construction

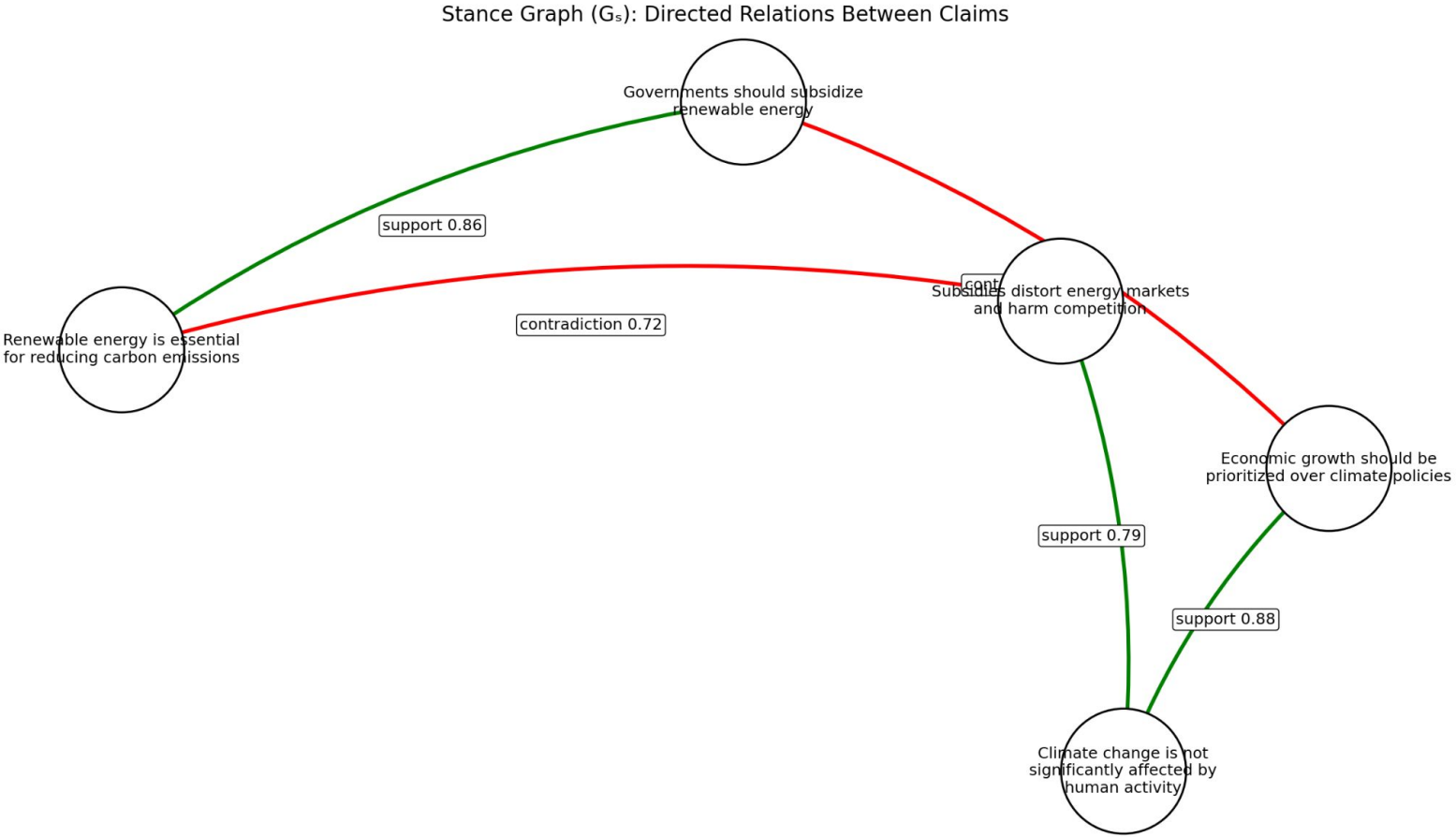
Extracted claims are represented as vertices in a directed graph. High-confidence stance relations form directed edges annotated with relation type and probability.

Two graphs are constructed:

- **Stance Graph (G□)**: Used for structural reasoning.
- **Logical Graph (G□)**: Enriched with fallacy labels and metadata.

This representation allows reasoning at both **local (edge-level)** and **global (graph-level)** scales.

Generated Stance Graph



Claim-Level Fallacy Classification

A DistilRoBERTa-based classifier is fine-tuned on the causalNLP/logical-fallacy dataset to assign fallacy labels to individual claims.

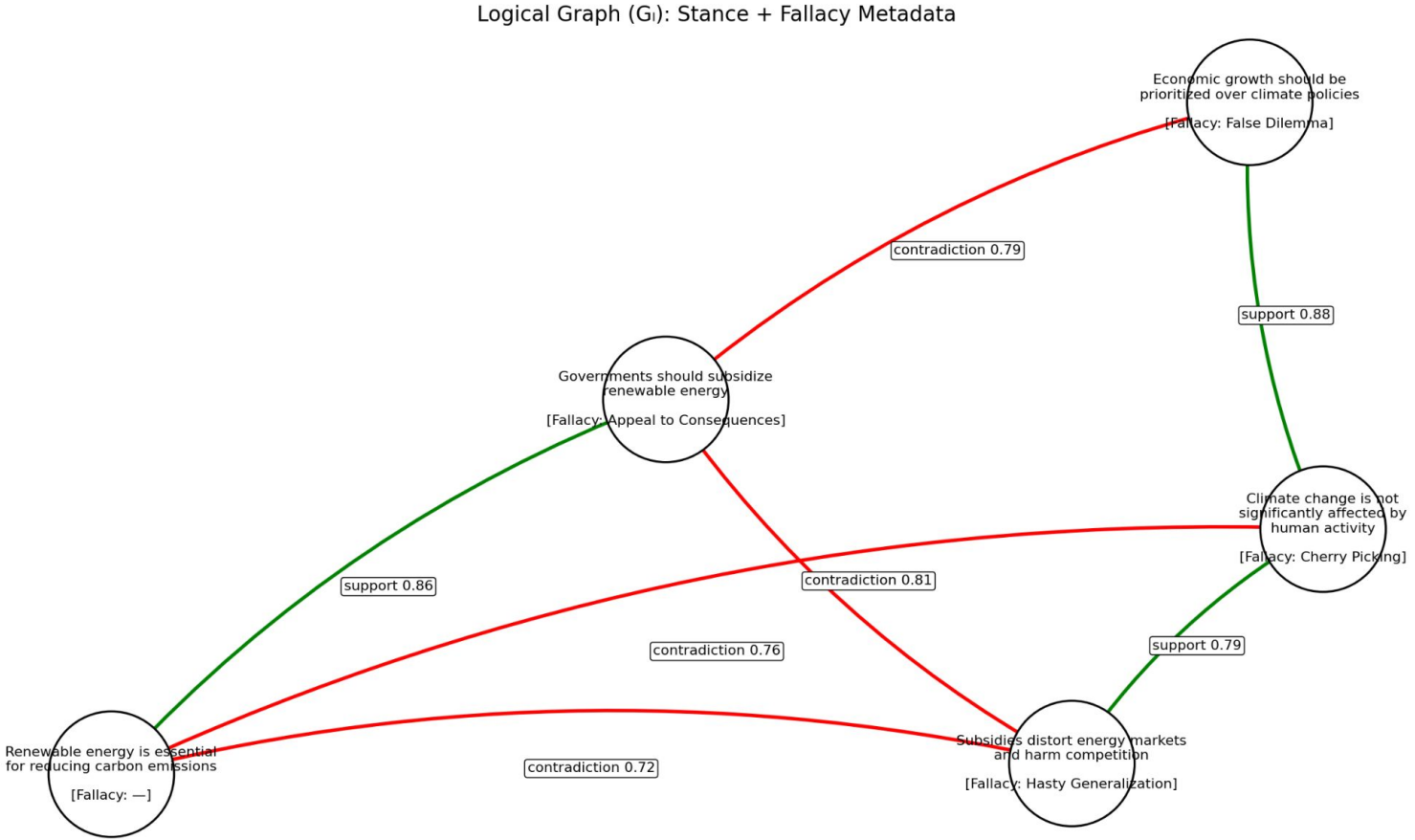
Training Setup

- Token length: 256
- Optimizer: AdamW (2e-5)
- Epochs: 2
- Loss: Cross-entropy

Results

- Macro-F1 improved from **0.20 → 0.31**
- Correctly identifies circular reasoning, false causality, and credibility fallacies.
- Provides complementary signals to graph-based reasoning.

Generated Logic Graph



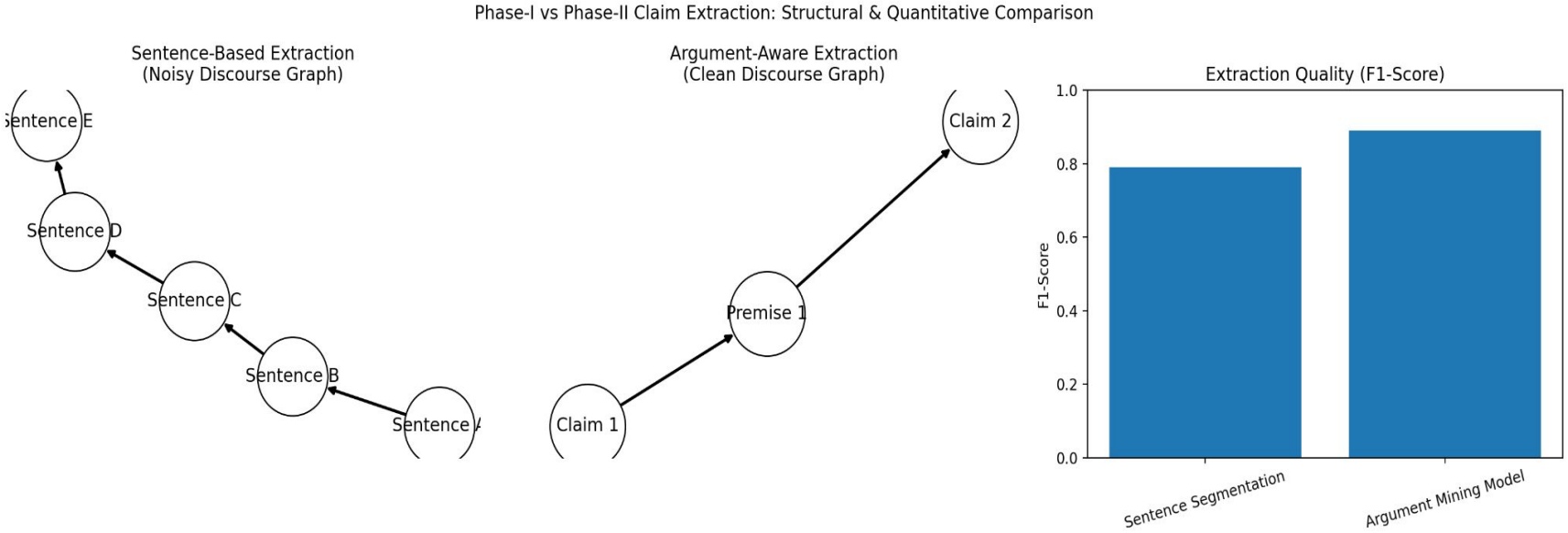
Argument-Aware Claim Extraction

Initial Phase-I implementations relied on sentence-level segmentation, which often included descriptive or narrative text that did not contribute to argumentative reasoning. In Phase-II, this was improved by integrating a **supervised argument mining model** capable of distinguishing argumentative statements from non-argumentative text.

The model classifies text spans into argumentative units such as claims and premises, ensuring that only semantically meaningful statements are introduced as nodes in the discourse graph.

Method	Precision	Recall	F1-Score
Sentence segmentation	0.82	0.76	0.79
Argument mining model	**0.91**	**0.88**	**0.89**

Argument-Aware Claim Extraction



Stance Detection Using Debate-Tuned NLI Mc

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Stance detection determines whether one claim supports, contradicts, or is neutral with respect to another claim. In Phase-II, the RoBERTa-MNLI model was **fine-tuned on debate-specific datasets** to improve its sensitivity to argumentative relations.

Each ordered claim pair is evaluated, and only high-confidence predictions are retained to form directed edges in the stance graph.

Model Configuration

- Base architecture: RoBERTa-large
- Task formulation: Natural Language Inference
- Labels: support, contradict, neutral
- Confidence thresholding applied to edge creation

Stance Detection Benchmark Results

The improved stance detector was evaluated on debate-oriented datasets and compared against baseline NLI performance.

Model	Accuracy	Precision	Recall	F1
Generic RoBERTa-MNLI	0.26	0.43	0.33	0.1841
Semeval2016 + DeBERTa-v3-large-mnli with Logistic Regression (Training Logistic Regression Model)	**0.64**	**0.62**	**0.67**	**0.63**

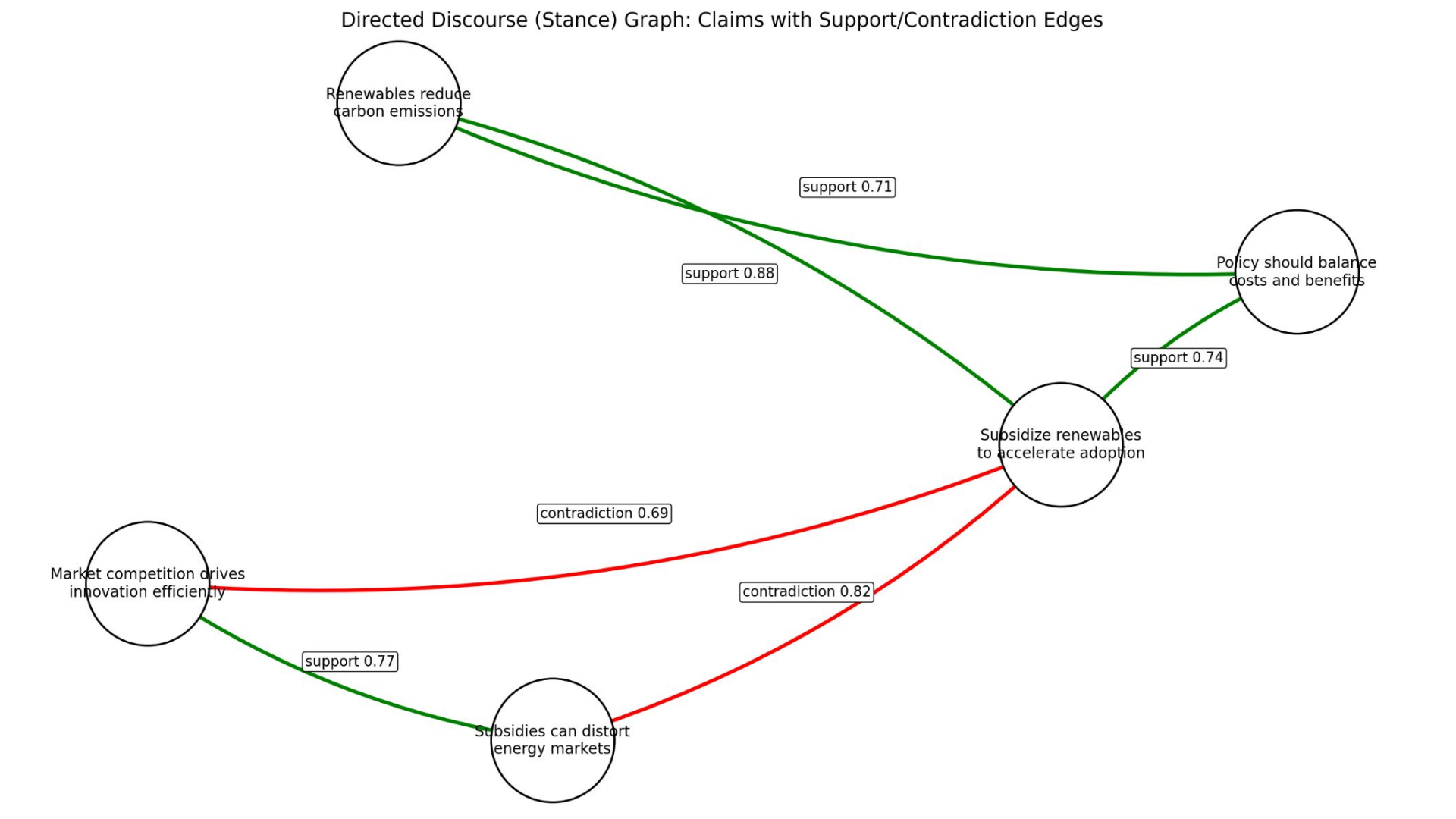
The fine-tuned model shows a clear improvement in identifying contradiction and support relations, which is critical for reliable graph-based reasoning.

Discourse Graph Representation

Extracted claims are represented as nodes in a directed graph, while stance relations form labeled edges. The resulting **stance graph** captures the argumentative structure of the debate at a global level.

Graph construction enables:

- Explicit representation of support and contradiction
- Structural reasoning across multiple claims
- Detection of emergent reasoning patterns not visible at sentence level


Directed Discourse (Stance) Graph: Claims with Support/Contradiction Edges
Renewables reduce carbon emissions
Policy should balance costs and benefits
Subsidize renewables to accelerate adoption
Market competition drives innovation efficiently
Subsidies can distort energy markets
support 0.71
support 0.88
support 0.74
contradiction 0.69
contradiction 0.82
support 0.77

Structural Fallacy Detection Using Graph Algorithms

Graph algorithms are applied to identify reasoning flaws that emerge only at the discourse level.

Implemented Techniques

- **Contradiction Detection**: Identifies explicit contradicting edges.
- **Cycle Detection**: Uses directed cycle enumeration to detect circular reasoning.
- **Support In-Degree Analysis**: Flags weakly supported claims.

Key Findings

- Detected multiple high-confidence contradiction loops.
- Claims labeled as circular reasoning frequently appear inside detected cycles.
- Structural and textual fallacy signals show strong alignment.

Learning-Based Fallacy Classification

Each claim is passed through a transformer-based fallacy classifier trained on annotated logical fallacy datasets. The classifier assigns a fallacy label and confidence score to each claim, which is later combined with graph-level reasoning.

Training Setup

- Model: RoBERTa-based classifier
- Loss: Multi-class cross-entropy
- Data: Logical fallacy annotated corpora

Model	**Macro-F1**
Baseline classifier	0.31
Enhanced classifier	**0.42**

Graph-Based Contradiction and Circular Reasoning Detection

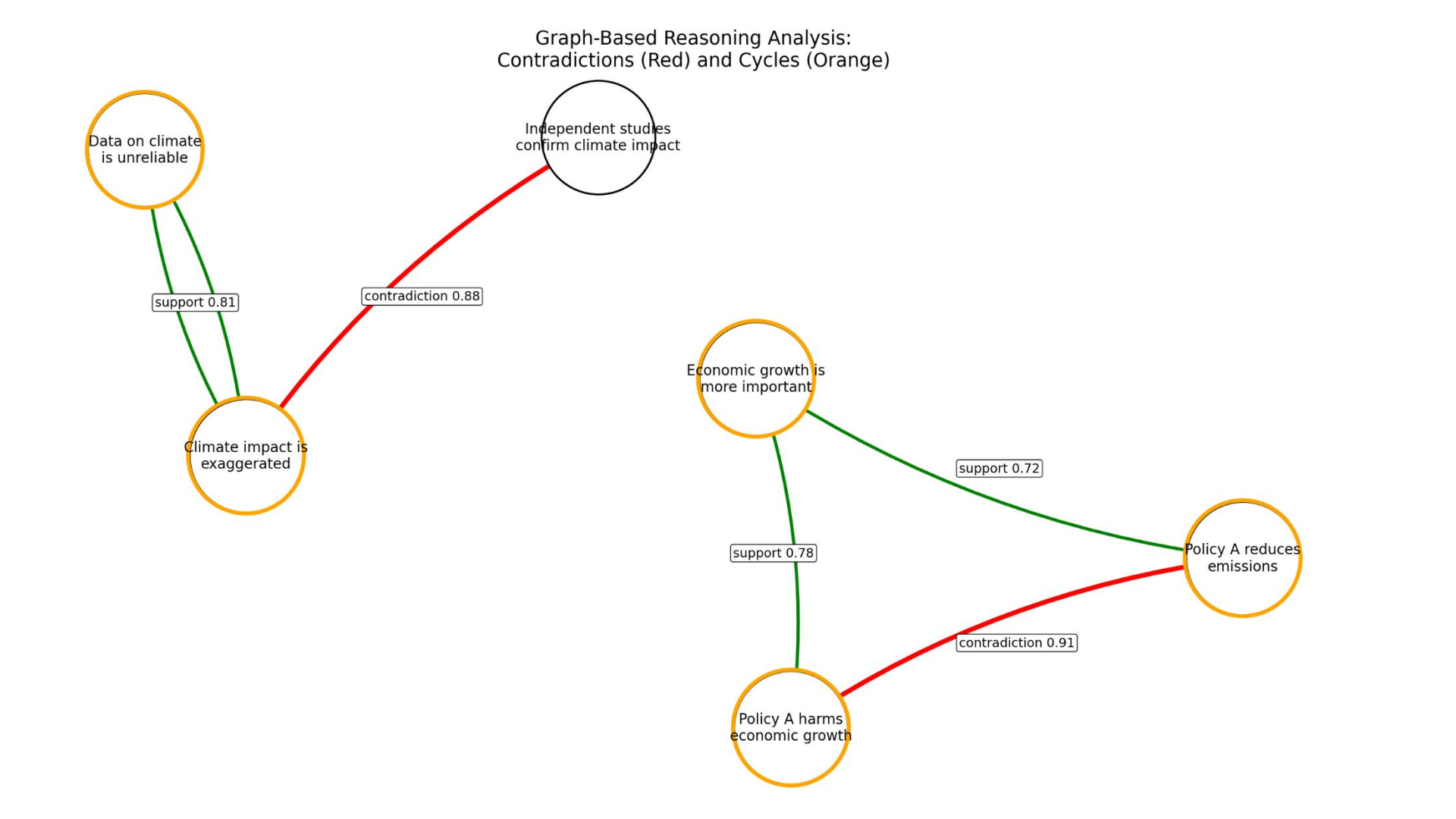
Graph algorithms are applied over the stance graph to detect reasoning flaws that span multiple claims.

Implemented Analyses

- **Contradiction detection:** Identification of explicit contradicting edge
- **Cycle detection:** Discovery of directed cycles indicating circular reasoning
- **Support analysis:** Identification of weakly supported or isolated claims

Observed Results

Pattern	Count
High-confidence contradictions	31
Circular reasoning cycles	14
Weakly supported claims	39


Graph-Based Reasoning Analysis:
Contradictions (Red) and Cycles (Orange)
Data on climate is unreliable
Independent studies confirm climate impact
support 0.81
contradiction 0.88
Climate impact is exaggerated
Economic growth is more important
support 0.72
support 0.78
Policy A reduces emissions
contradiction 0.91
Policy A harms economic growth

Efficient Graph Construction at Scale

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Naive stance detection scales quadratically with the number of claims. To address this, semantic pruning was introduced to eliminate claim pairs that are unlikely to be related.

Using sentence embeddings and similarity thresholds, irrelevant pairs are filtered before NLI evaluation.

Results

- 5×–10× reduction in runtime on large debates
- Minimal loss of meaningful support or contradiction edges
- Improved feasibility for real-world discourse analysis

DebateIdx	VertexCount	EdgeCount	OldTime(s)	NewTime(s)	Speedup
0	15	210	186.49	15.62	11.94×
1	27	992	976.67	117.62	8.30×
2	26	650	746.06	84.55	8.82×
3	16	272	142.84	19.10	7.48×
4	34	1122	503.69	76.14	6.61×

Scalability Analysis and Semantic Pruning

Initial graph construction required $O(n^2)$ stance evaluations, leading to excessive runtime for large debates.

To address this, semantic pruning using Sentence-BERT embeddings was introduced. Claim pairs with cosine similarity below 0.3 were discarded prior to NLI evaluation.

Results

- Achieved **5×–10× runtime speedup** on EthiX dataset.
- Reduced graph density without significantly affecting contradiction detection.
- Demonstrates feasibility for large-scale discourse analysis.

Bottleneck in our previous work

Our results in the previous review was bottlenecked by the performance of the stance prediction model.

Although the graph modelling and algorithms were put in place, the overall system did not work for complex graphs/debates.

Thus we have focused on improving the stance prediction model before moving on to further graph analysis algorithms.

Papers we are following

There are 2 main papers we are following to improve our stance model,

1. GraphNLI: A Graph-based Natural Language Inference Model for Polarity Prediction in Online Debates

   - Implements a graph based ML algorithm to perform stance detection, includes argument context by graph traversal. Very relevant to our problem statement of performing stance detection on graphs.

2. Extractive Fact Decomposition for Interpretable Natural Language Inference in One Forward Pass

   - Demonstrates the effectiveness of Claim decomposition from arguments in increasing the accuracy of stance detection

Models we have benchmarked for stance detection

1. Roberta-large-mnli: Was used in our previous previous, will server as a low baseline performance bar.

2. Stance-detection-semeval2016: Pre-Trained on the SemEval-2016 Task 6 to perform stance detection

3. DeBERTa-v3-large-mnli: Pre-Trained on MultiNLI dataset.

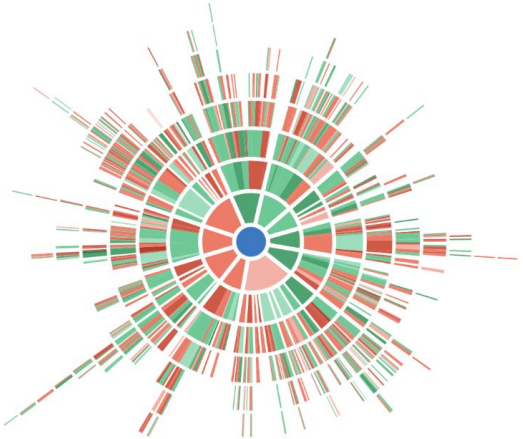
4. Semeval2016 + DeBERTa-v3-large-mnli with Logistic Regression (Training Logistic Regression Model)

Dataset used

- Due to the unique constraints of the problem(graph based debate modelling with stance detection), there are no publically available datasets to benchmark.
- Although the Kialo dataset used by **Graph-NLI** models debates as a DAG(Directed Acyclic Graph) and can be used for stance detection, it is not publically available.
- Due to the constraint of that dataset, we have used synthetic Kialo-based generated dataset **syncIALO** to benchmark all the models.

Dataset Structure of **synclALO**

Total number of train debates: 1000
Total number of test and val debates: 50
Total number of claim-claim pair for stance detection: 560,000



(b) Visualization of this debate's tree



(a) An example of the arguments made in a Kialo debate. The thesis is, "*Pregnant people should have the right to choose abortion*". A supporting reply is, "*Access to legal abortion improves the health and safety of women*". An attacking reply to this support is, "*When abortion is easily available, it incentivises irresponsible behaviour*".

Benchmark Results from Roberta-large-mnli (prev review model)

	precision	recall	f1-score
support	0.9677	0.0617	0.1161
attack	0.2672	0.9374	0.4158
neutral	0.0682	0.0120	0.0204
accuracy			0.2605
macro avg	0.4344	0.3370	0.1841
weighted avg	0.3376	0.2605	0.1446

Overall performance

- **Accuracy: 26.1%** (very low)
- The stance classifier shows strong bias toward ATTACK(high recall), achieving high recall at the cost of precision and fails to model SUPPORT as seen by the low recall and high precision.
- As the initial model used, this has the worst performance

Benchmarking on Stance-detection-semeval2016

	precision	recall	f1-score
support	0.2439	0.5329	0.3346
attack	0.2756	0.1526	0.1965
neutral	0.5710	0.3729	0.4511
accuracy			0.3555
macro avg	0.3635	0.3528	0.3274
weighted avg	0.4160	0.3555	0.3578

Overall performance

- **Accuracy: 35.5%** (2x prev model)
  Macro f1: 32% (2x prev model)
- The model over predicts support as seen by high recall and low precision, model also does not detect recall well as seen by low recall.

Benchmark on DeBERTa-v3-large-mnli

support P=1.00 R=0.035 F1=0.068 (486)
attack P=0.37 R=0.824 F1=0.511 (511)
neutral P=0.45 R=0.381 F1=0.413 (1003)

accuracy = 0.410 (6%+ from prev model)
macro_F1 = 30%

The MNLI models predict attack frequently while the sem_eval stance model has the opposite problem.

Semeval2016 + DeBERTa-v3-large-mnli with Logistic Regression (Training Logistic Regression Model)

```
Accuracy: 0.6252072968490879
[[218  17  62]
 [ 20 237  44]
 [155 154 299]]
              precision    recall  f1-score

     support     0.5547    0.7340    0.6319
      attack     0.5809    0.7874    0.6685
     neutral     0.7383    0.4918    0.5903

    accuracy                         0.6252
   macro avg     0.6246    0.6711    0.6303
weighted avg     0.6538    0.6252    0.6201
```

Since DeBRETA was better at classifying attack and Semeval on support we combine the both by feeding their logits to a logistic regression model.

Results:
Accuracy: 62% (20%+ prev model)
Macro_f1: 63% (30%+ prev model)

This method also balanced out precision and recall across support and attack, proving that by using stance + nli model we were able to balance their disadvantages,

SOTA Results (no LLM)

SOTA Results are held by LLM models (90%+ accuracy).
Non-LLM SOTA in stance detection in graph based modelling of debates is held by Graph-NLI.

To improve accuracy we will have to train explicitly on this dataset by using graph walks to provide context to stance model similar to GraphNLI.

Model	Accuracy (%)
GraphNLI: Root-seeking Graph Walk + Sum	80.70
GraphNLI: Root-seeking Graph Walk + Avg.	81.96
GraphNLI: Root-seeking Graph Walk + Weighted Avg.	**82.87**
GraphNLI: Biased Root-seeking Random Walk + Sum	79.95
GraphNLI: Biased Root-seeking Random Walk + Avg.	80.44

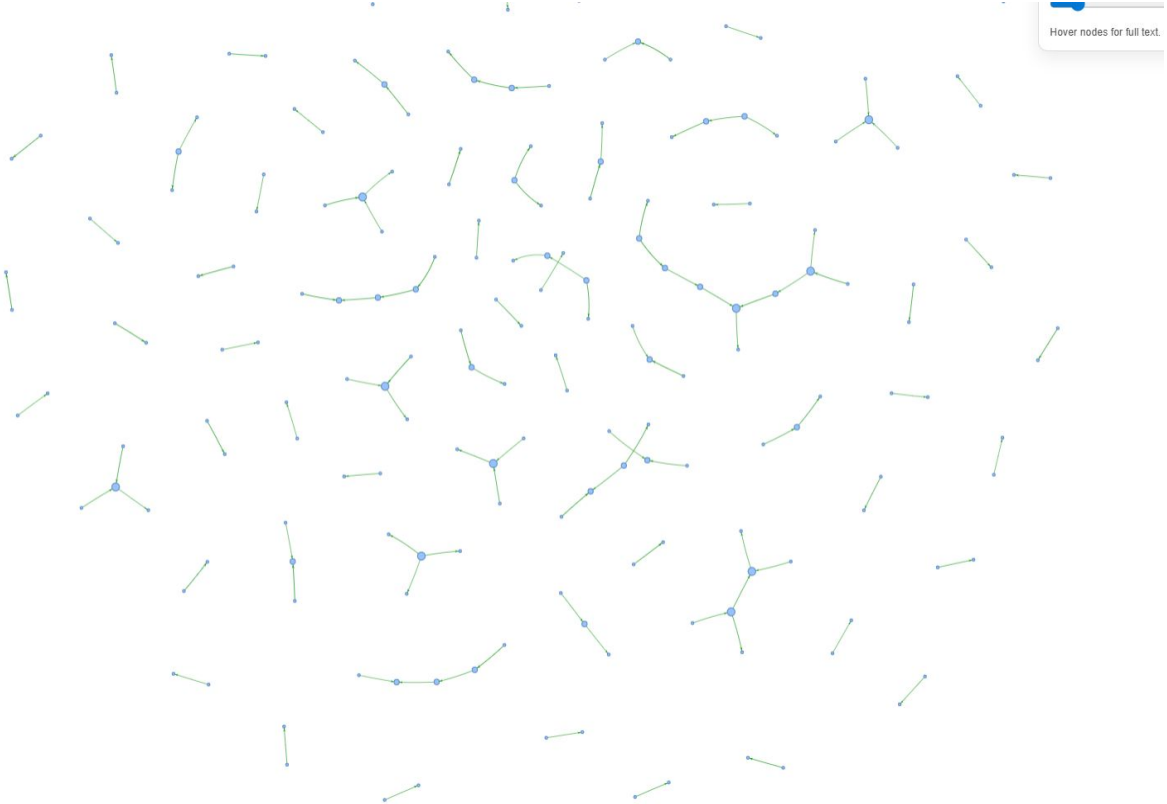
Graph Creation

Graph Creation is being done by using the stance predictions from the stance prediction model(Semeval2016 + DeBERTa-v3-large-mnli with Logistic Regression).

Nodes: Claim from user
Edge: Stance between nodes/claims

1. Stance Model is evaluated on every pair of claims/nodes.
2. The resultant model output classifies their relationship into {support, neutral, attack}, this value is given to the edge connecting the pair of nodes.

Support Graph



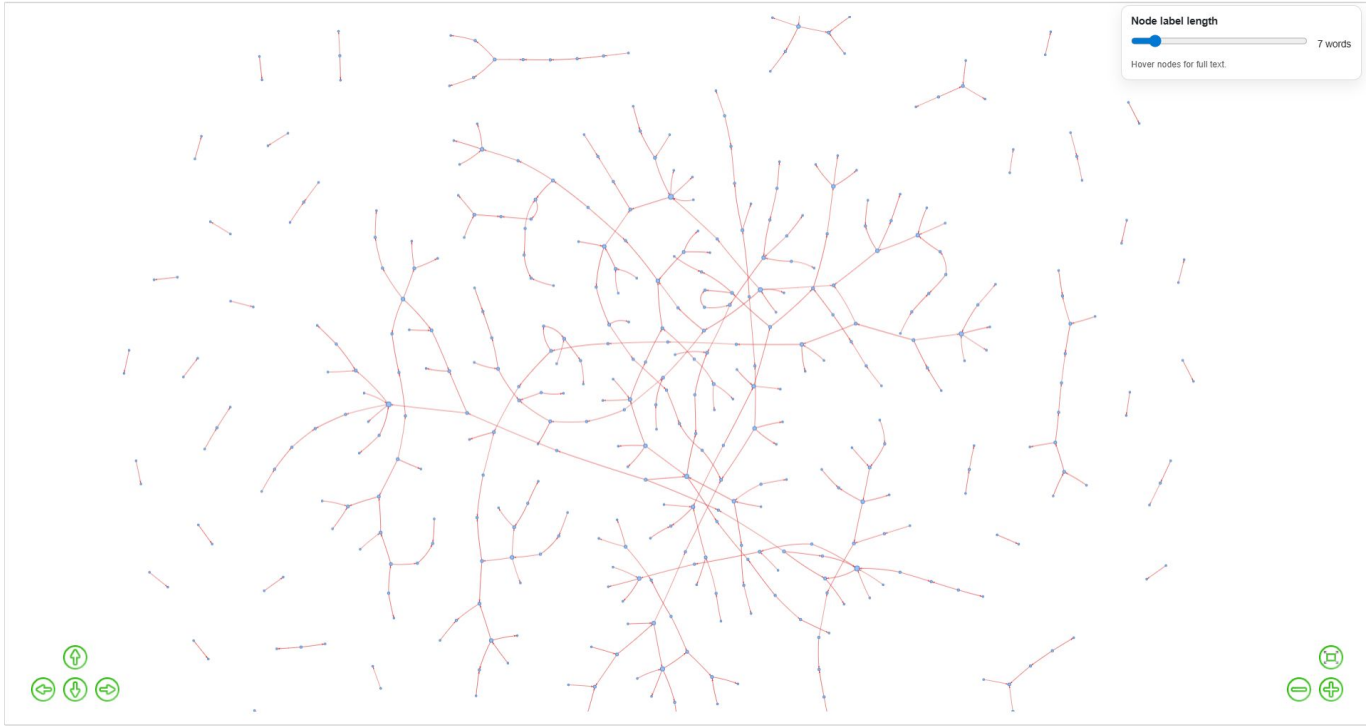
Model: Semeval2016 + DeBERTa-v3-large-mnli with Logistic Regression

Dataset: **syncIALO**

Debate_idx = 2

edges(u_i,u_j) ∈ {1}

Attack Graph



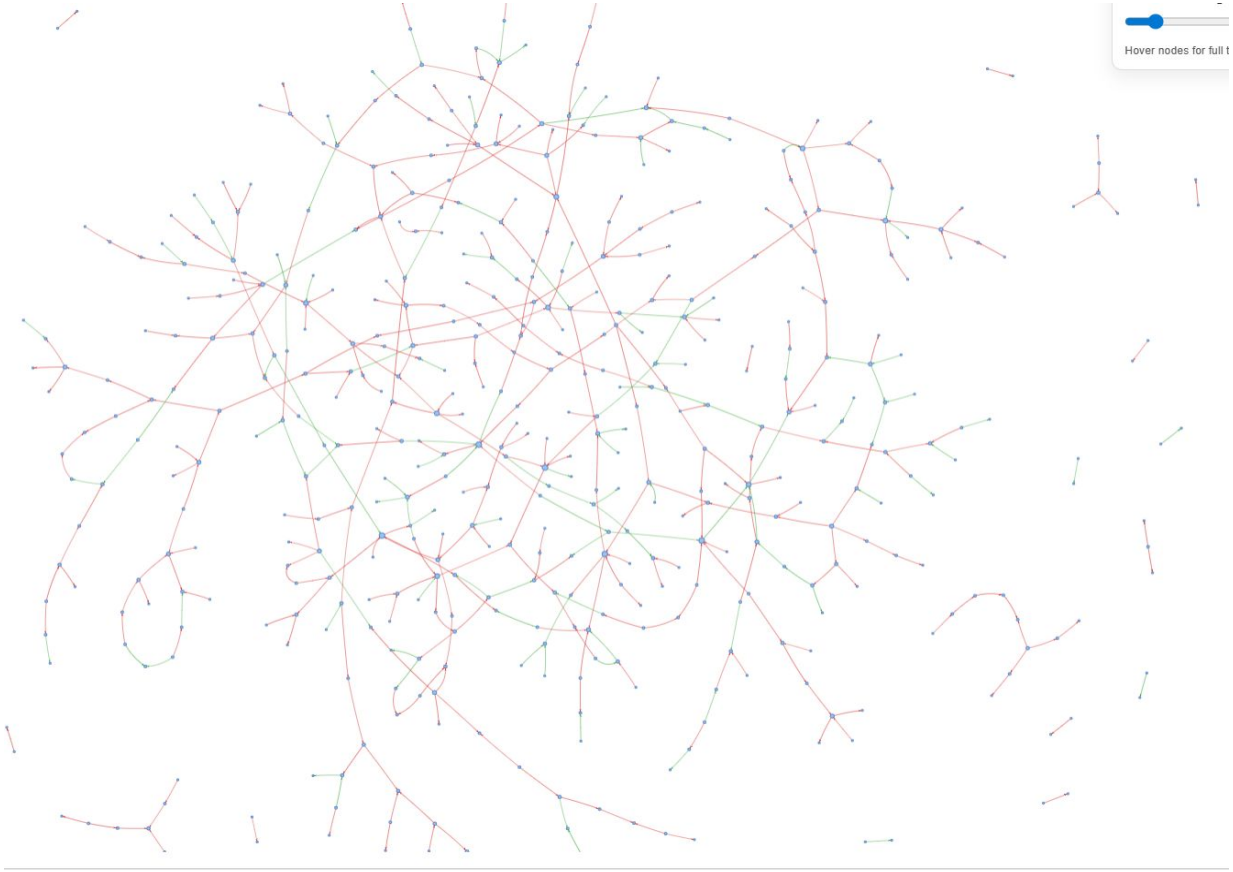
Model: Semeval2016 + DeBERTa-v3-large-mnli with Logistic Regression

Dataset: **synclALO**

Debate_idx = 2

edges(u_i,u_j) ∈ {-1}

Combined Graph (attack+support)



Model: Semeval2016 + DeBERTa-v3-large-mnli with Logistic Regression

Dataset: **syncIALO**

Debate_idx = 2

edges(u_i,u_j) ∈ {1,-1}

Graph Algorithms for Automatic Analysis

We aim to utilize various Graph Algorithms to analyze the structure of large scale debates.
Some of the algorithms we are using,

1. PageRank: To analyze the **support and attack structure** of nodes from a global perspective(unlike in_degree which only has a local perspective)

2. Minimum-Cut: - Min-Cut measures the **vulnerability of a support graph/sub-graph**. It find the minimum amount of claims that have to be disproven or attacked to break the support structure and cohesive argument made in the sub-graph.

3. Graph Clustering: Clustering will help us identify the overall structure of the debate claims overall structure of the debate, from **finding the sub-topics** discussed in each cluster to the **evolution of the overarching ideas** in the debate with the advance of time.

Algorithm 1 PageRank

Require: Directed graph $G = (V, E)$, damping factor β, tolerance ϵ
Ensure: PageRank vector PR

$N \leftarrow |V|$
Initialize $PR(u) \leftarrow 1/N \quad \forall u \in V$
repeat
  for each node $u \in V$ **do**
    $PR_{\text{new}}(u) \leftarrow \frac{1-\beta}{N}$
    for each edge $v \to u$ **do**
      $PR_{\text{new}}(u) \leftarrow PR_{\text{new}}(u) + \beta \cdot \frac{PR(v)}{L(v)}$
    end for
  end for
  $\delta \leftarrow \|PR_{\text{new}} - PR\|_1$
  $PR \leftarrow PR_{\text{new}}$
until $\delta < \epsilon$
return PR =0

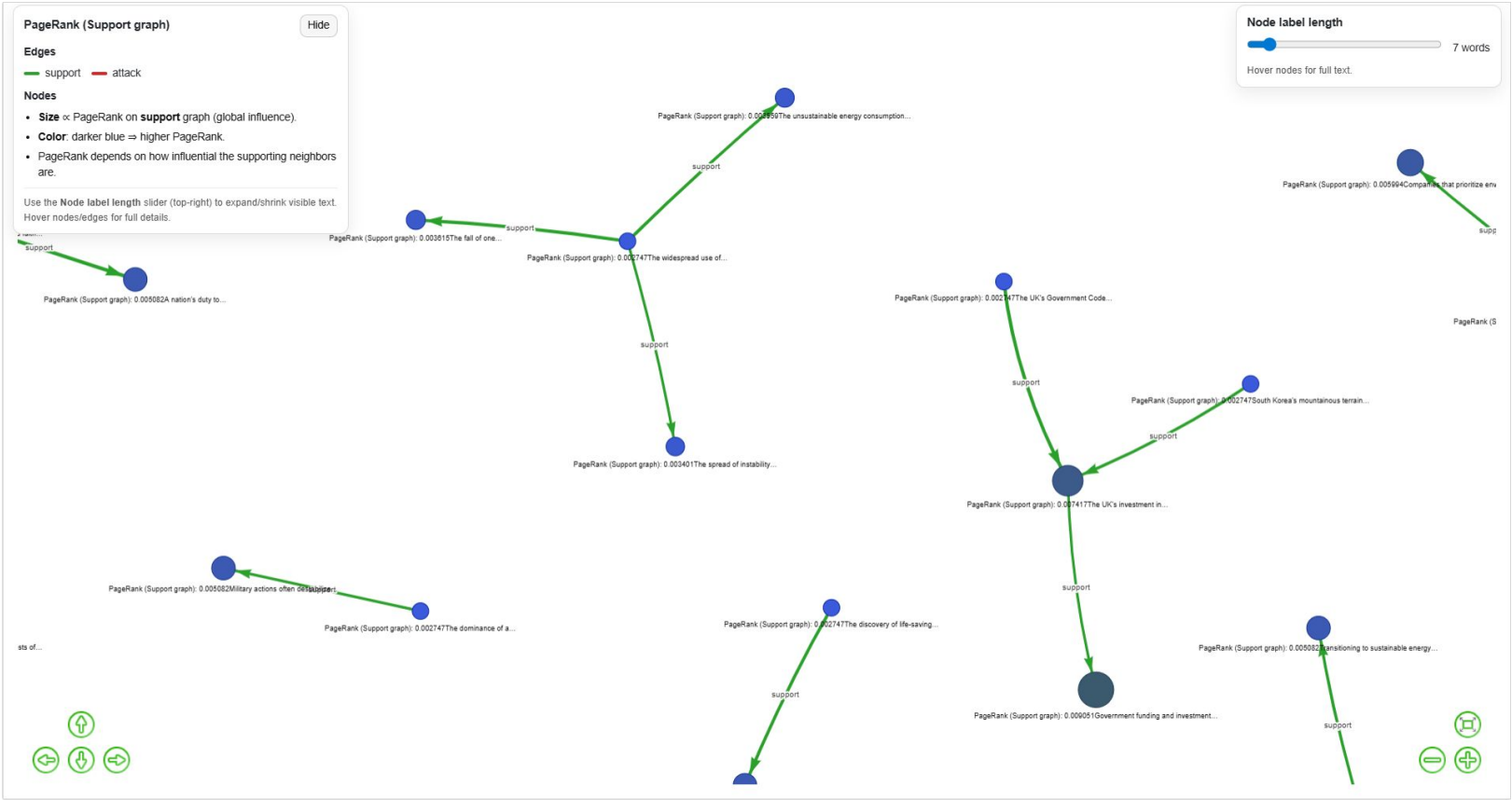
In our scenario, the page rank algorithm is done on 3 graphs- exclusive support, exclusive attack and combined support+attack

For the exclusive support+attack graph we find the net Page Rank for each node being: PageRank_{support}(u) - PageRank_{attack}(u)

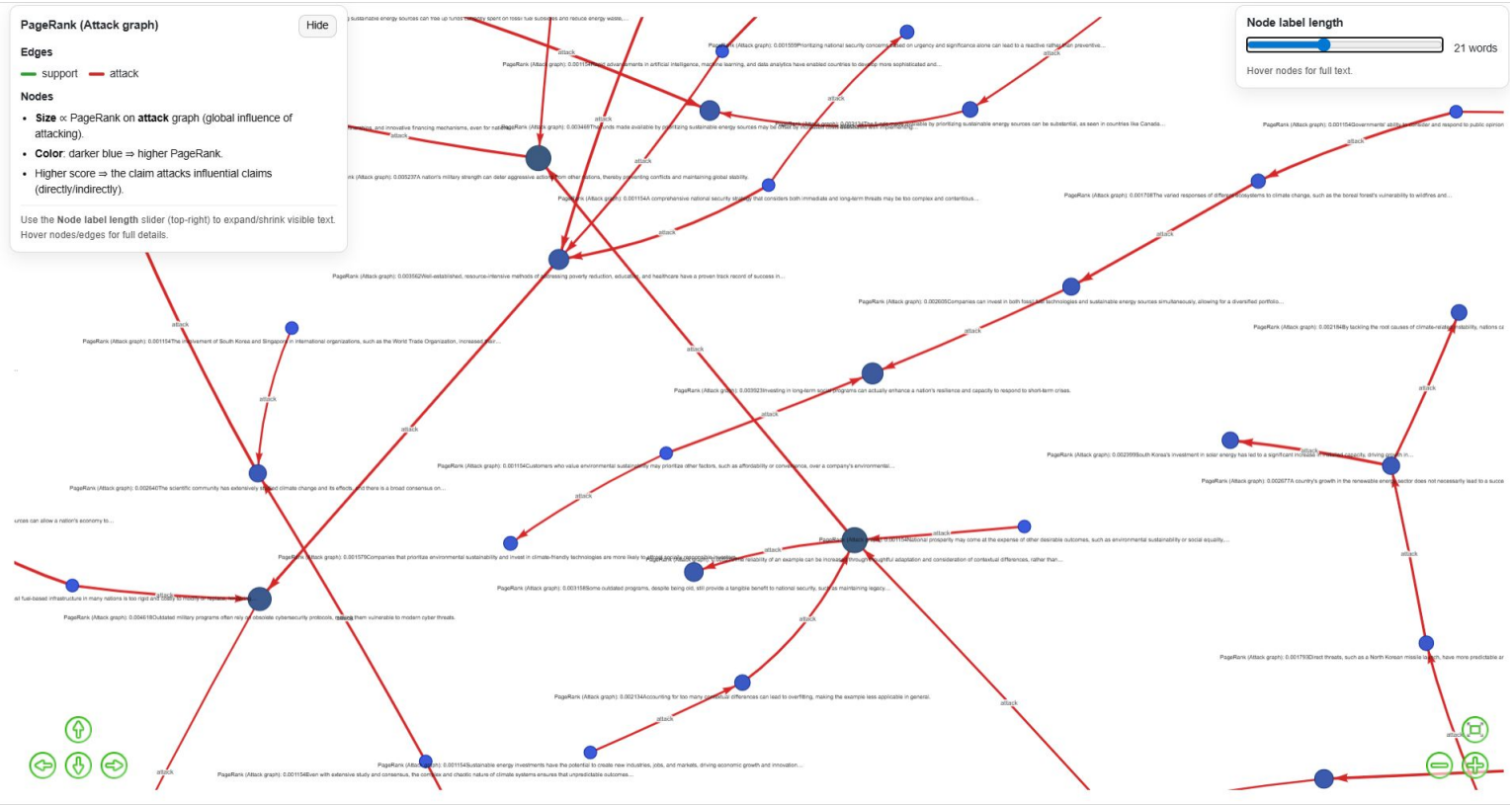
Page Rank of a support graph gives you the globally most well supported claim/node.

Similarly PageRank of an attack graph gives you the globally most attacked claim/node.

Results of PageRank for support graph

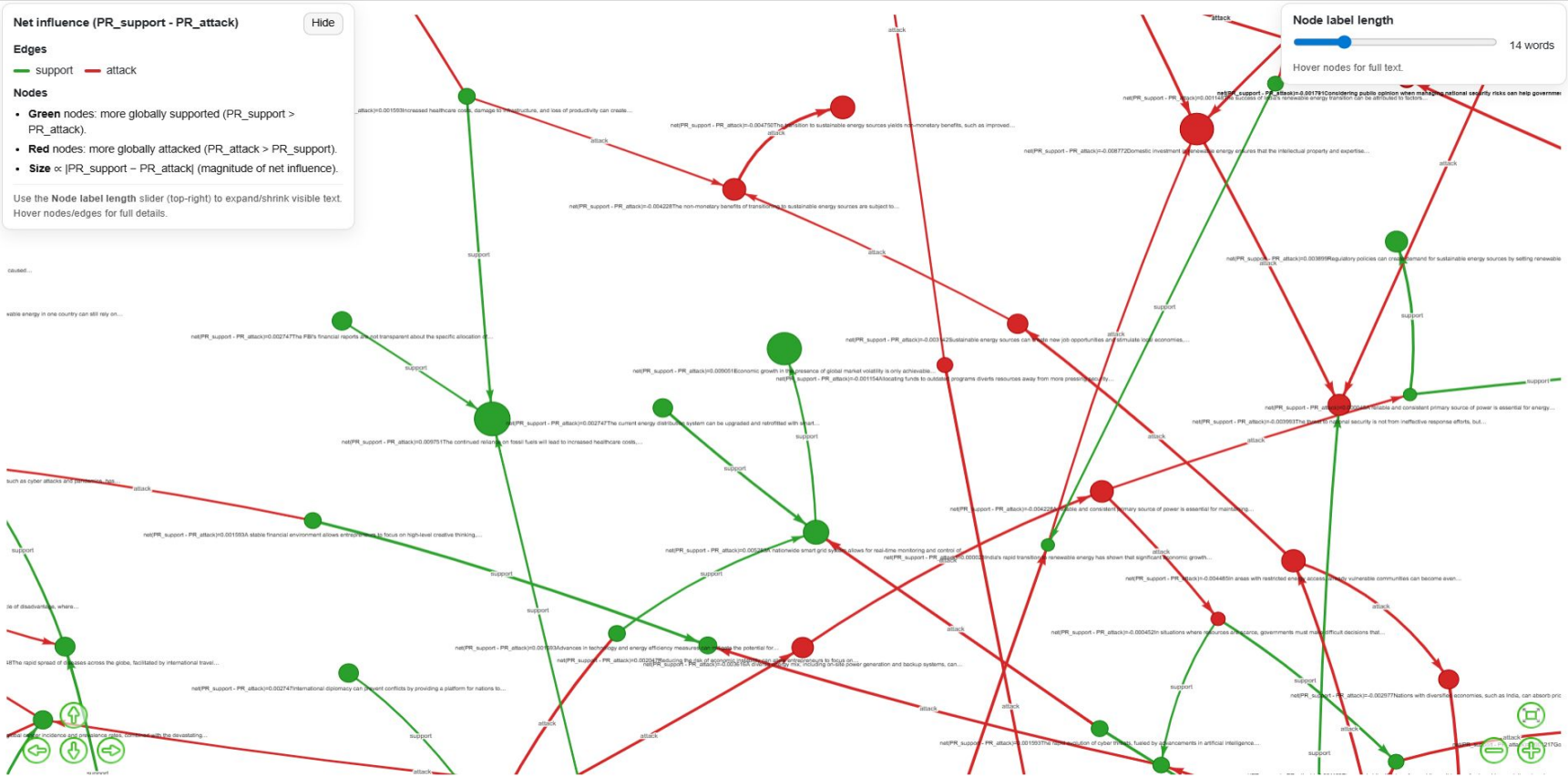


Results of PageRank for attack graph

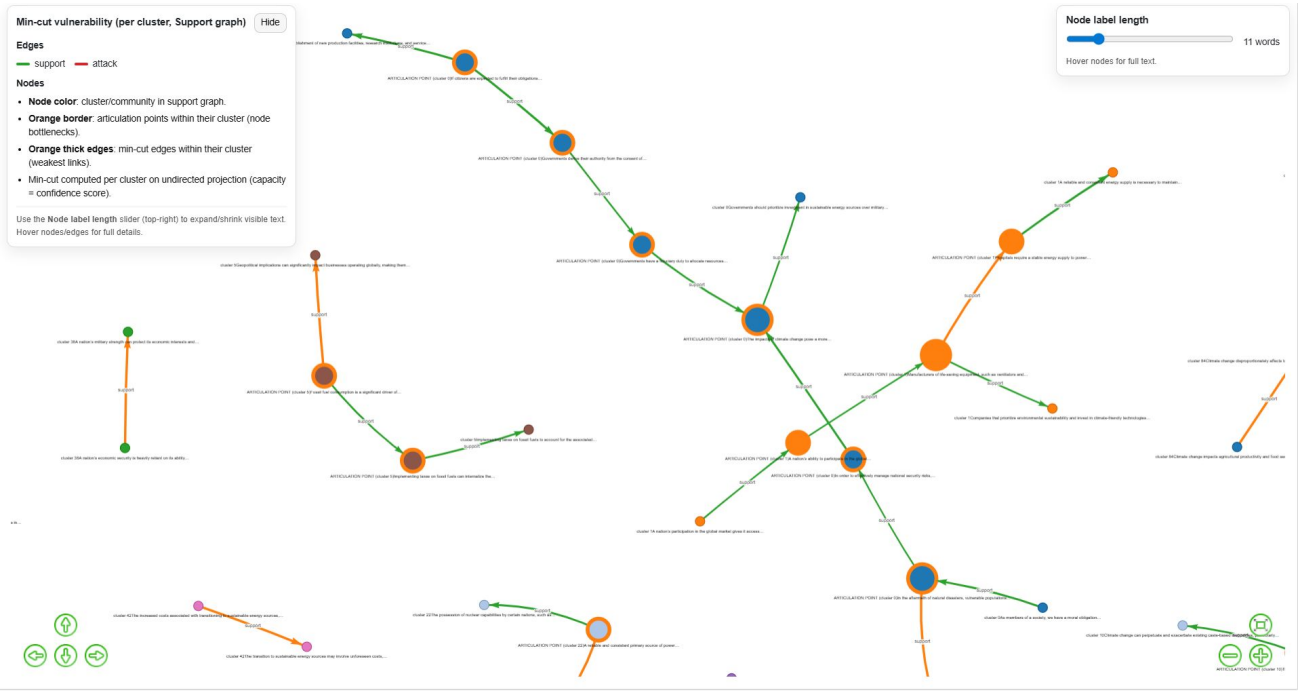


Larger size of node indicates larger PageRank value

Results of Page Rank on Combined Graph



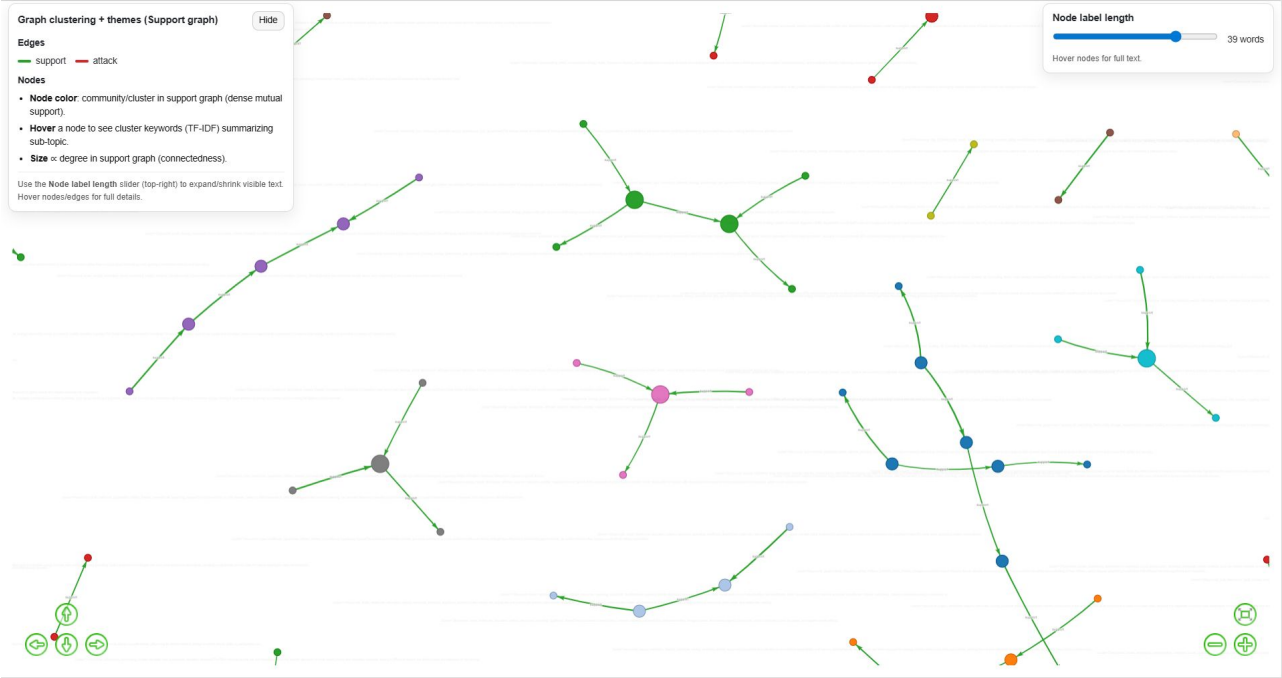
Results from Min Cut Algorithm



Orange nodes indicate claims that break a graph into 2/more subgraphs if removed. (weakest node)

Orange Edge indicates the actual min cut edges in the subgraph.

Results of Graph Clustering



Nodes of the same colour belong to the same cluster.

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